



Vocal Complexity in Meerkats and Other Mongoose Species

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1. INTRODUCTION

1.1. Social Complexity and Communicative Complexity

The influence of a species' social environment on its communication has been acknowledged by Darwin and even earlier on (for a recent review, see Freeberg, Dunbar, & Ord, 2012), but has only recently become a

major focus of research from insects to mammals including humans (Blumstein & Armitage, 1998). The hypothesis is that socially more complex species also show more complex ways of communication, yet currently, little empirical evidence solidly supports this claim. This may be due to the lack of clarity on what “complexity” means in terms of both the social and communicative aspect. In two recent papers, a move toward more clear terminology has emerged. Freeberg and colleagues (2012) clearly defined both of these terms, while Pollard and Blumstein (2012) outlined the possible functional relationship between specific attributes of social complexity and attributes of communicative complexity. Such clear predictions will help to test the influence of biologically relevant social attributes on the evolution of different communication systems and potentially allow us to explain the selective drivers and constraints inducing the communicative variation we observe in animals, including humans.

Complex social systems were defined by Freeberg and colleagues (Freeberg et al., 2012, p. 1787) as relating to “. . . systems in which individuals frequently interact in many different contexts with many different individuals, and often repeatedly interact with many of the same individuals over time.” This definition uses broad categories to encapsulate the social environment including unit size, unit density, member roles, social structure as well as interunit density as attributes to quantify the degree of social complexity. In their definition unit often refers to “group,” but could also mean population, subgroup, or whatever category is used as point of comparison. Pollard and Blumstein (2012) discuss in more detail the influence of group size, social bond strength, demographic role variability, reproductive role variability, and the mating system on specific communicative attributes. Some of these attributes overlap with Freeberg and colleagues’ broad categories or define them more narrowly. To enable comparative studies, it is crucial that the data on the quantified attributes are available for several different species, and to date this is seldom the case, or only for two to three species to compare.

Complex communicative systems were suggested to relate to those systems “that contain a large number of structurally and functionally distinct elements (e.g., large display repertoire sizes) or possess a high amount of bits of information” (Freeberg et al., 2012, p. 1787). While both of these attributes vary substantially among (and even within) species and thus represent a good potential proxy for communicative complexity, they may not always be easy to quantify. Distinct signals occur under some conditions, but very

often signals exist in graded forms, and without understanding the meaning for the receiver and identifying the boundaries between signal types, we may misinterpret the true function (Keenan, Lemasson, & Zuberbühler, 2013). Similarly, the calculation of the amount of bits of information is based on the assumption that the greater the variation of elements in a signaling system, the higher the potential for information or complexity within the system (Freeberg et al., 2012). Greater information could lead to more meaning in a system, but this may also depend on the cognitive abilities of a species to process this information. Only by testing how meaningful the observed variation in signals is to receivers can we begin to make final conclusions on the complexity of a communicative system. Nevertheless, to begin with, we may have to be content with the description of signal variation and what context it relates to.

The difficulty in identifying a relationship between social complexity and communicative complexity is not only due to the many different attributes within the social and communicative domain but also that often several of them may correlate and influence each other. In addition, besides social attributes, ecological attributes have been shown to play an important role in the evolution of the variation in communication (Freeberg et al., 2012).

1.2. Ecological Constraints on Communication

Besides social factors affecting communication in different species, the ecological environment, including physical and biotic aspects, needs to be taken into consideration when explaining the observed variation. It is clear how the physical properties of the habitat may favor the use of specific modalities, for example, in water or densely vegetated terrestrial habitat visual cues are not very effective and acoustic signals may be more important (Bradbury & Vehrencamp, 2011). However, signal evolution within the same modality may also be strongly influenced by the ecological environment thus impacting on the complexity of a species' communicative system. For example, the brightness of visual signals, or amplitude and frequency range of vocal signals must be adapted to transmit optimally within "noisy" environments. Prey animals also need to balance the needs of reduced conspicuousness to avoid predation versus effective communication with the intended receivers (e.g., vocal signals in birds, Haskell, 1999 and visual signals in fish, Magurran, Seghers, Carvalho, & Shaw, 1993). Biotic factors can furthermore act as a driver of increased repertoire size, such as large alarm call repertoires in environments with many predator types (Furrer & Manser,

2009a, 2009b; Macedonia & Evans, 1993) or specific food calls for different food types (Clay, Smith, & Blumstein, 2012; Hauser & Marler, 1993).

The difficulty in identifying a relationship between social, ecological, and communicative complexity can be ascribed to the numerous attributes that distinguish each domain, as well as the sometimes unanticipated interactions that various aspects of these domains may have on each other. For example, if two species differ slightly: with one being more socially complex, but the other living in more ecologically complex environments, to which domain can you attribute variation in communicative complexity? Only by comparing closely related species with similar ecological needs, but variance in social structure, or vice versa, can we really disentangle the relative roles of the social and ecological environment on communicative complexity (e.g., Gustison, le Roux, & Bergman, 2012). Currently, such a comparative approach is only possible for the vocal communication repertoires of a few bird (Ord & Garcia-Porta, 2012) and mammalian (Blumstein & Armitage, 1998; delBarco-Trillo, Sacha, Dubay & Drea, 2012; Wilkinson, 2003) taxonomic classes.



2. MEERKATS AND OTHER MONGOOSE SPECIES

2.1. Study Species

We have studied five mongoose species of the monophyletic family of *Herpestidae* (Agnarsson, Kuntner, & May-Collado, 2010), including meerkats (*Suricata suricatta*), banded mongooses (*Mungos mungo*), dwarf mongooses (*Helogale parvula*), yellow mongooses (*Cynictis penicillata*), and slender mongooses (*Galerella sanguinea*). These mongoose species provide a unique opportunity to address questions regarding the influence of the social and the ecological environment on the evolution of communicative complexity within phylogenetic closely related species. They are highly vocal carnivores of small size that exhibit variation in social structure from solitary systems to obligate social groups. The communication systems of these five mongoose species have been systematically studied, but to different extents. Currently, extensive information exists for meerkats, while more limited data exist for banded mongooses, dwarf mongooses, yellow mongooses, and slender mongooses.

These five mongoose species occur in Africa and partly overlap in their geographical distribution (Fig. 6.1). They are of similar size, with dwarf mongooses representing the lower end of the spectrum at approximately

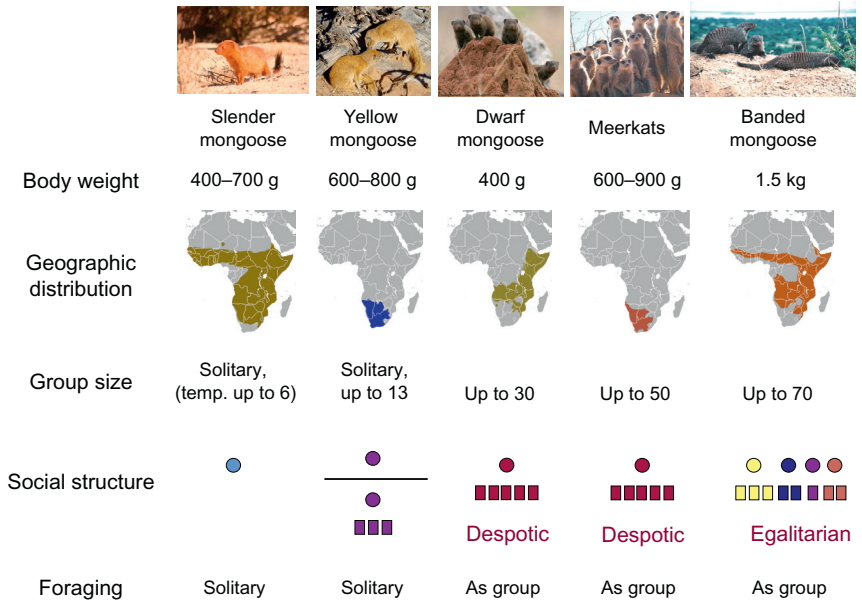


Figure 6.1 Comparison of five different mongoose species on geographic distribution, size, group structure and foraging behaviour. Slender Mongoose (*Galerella sanguinea*) range by Chermundy, available at http://en.wikipedia.org/wiki/File:Slender_Mongoose_area.png; Yellow Mongoose (*Cynictis penicillata*) range by Chermundy, available at http://en.wikipedia.org/wiki/File:Yellow_Mongoose_area.png; Common Dwarf Mongoose (*Helogale parvula*) range by Chermundy, available at http://en.wikipedia.org/wiki/File:Common_Dwarf_Mongoose_area.png; Meerkat (*Suricata suricatta*) range by Chermundy, available at http://en.wikipedia.org/wiki/File:Meerkat_area.png; Banded Mongoose (*Mungos mungo*) range by Chermundy, available at http://en.wikipedia.org/wiki/File:Banded_Mongoose_area.png. All five base maps derived from [File:BlankMap-World.png](#). Distribution data from *IUCN Red List*.

300 g, the banded mongooses at the upper end reaching 1.5 kg and the other three species (meerkats, slender mongooses, and yellow mongooses) in between 400 and 800 g adult body weight. All species differ slightly in the habitat they use, with meerkats and yellow mongooses occurring in more open, dry areas, whereas banded mongooses and dwarf mongooses live in more vegetated parts, and the slender mongooses overlap with the geographic distribution of the other four species. While such habitat variation gives rise to slight diet differences between these herpestids, all five species are exposed to a suite of predators (raptors, middle-sized terrestrial predators, and snakes) due to similarities in body size and foraging preferences. All of them share their habitat with nonherpestid species and are often seen to

interact with them, whether mutualistically (e.g., [Makenbach, Waterman, & Roth, 2013](#)) or competitively ([Flower, 2011](#)).

The five mongoose species live in different social systems and differ in group size, group structure, spatial cohesion while active and, in particular, while foraging, as well as in how they raise their offspring ([Fig. 6.1](#)). This variation affects the coordination of their group activities, for example, only the social obligate species forage as a cohesive unit, while the solitary slender mongoose and the facultative social yellow mongoose primarily forage alone. Whereas in the slender mongoose, the female rears her offspring by herself ([Waser, Keane, Creel, Elliott, & Minchella, 1994](#)), in the other four species, group members help in raising the offspring ([Schneider & Kappeler, 2014](#)). In the yellow mongoose ([Vidya, Balmforth, le Roux, & Cherry, 2009](#)), meerkat ([Clutton-Brock et al., 2001](#)), and dwarf mongoose ([Keane, Creel, & Waser, 1996](#); [Rasa, 1987](#)), we find a mainly despotic reproductive system with the dominant-pair monopolizing breeding, whereas in the banded mongoose reproduction is more egalitarian, and if they are in good condition all adult females and males can breed ([Cant, 2000](#); [Rood, 1975](#)). The duration of lasting social relations also vary between species: slender mongoose pairs do not remain together long enough to cooperatively rear offspring, yellow mongoose pairs may reunite year after year, with offspring sometimes remaining at the den for more than one breeding season; in contrast to the obligate social species in which no individual is ever outside of a social group for longer than a few days (prospecting trips).

2.2. Study Populations

The data on all five species have been collected in their natural habitat, with few references to earlier studies on captive groups. Most of the work on meerkats has been performed at the Kalahari Meerkat Project based at the Kuruman River Reserve (www.kalahari-meerkat-project.com) in the southern part of the Kalahari in South Africa (for detailed description, see [Clutton-Brock et al., 1998](#)). At the same study site, the data on the sympatric yellow mongoose and slender mongoose species have been collected. Banded mongooses have been studied in detail at the Banded Mongoose Project (www.bandedmongoose.org) at Queen Elizabeth National Park in Uganda (for detailed description, see [Cant, Vitikainen, & Nichols, 2013](#)). The collection of data on vocal behavior of meerkats, banded mongooses, and yellow mongooses is based on the same approach. Several different

groups in these populations have been habituated to a stage that allows observers to follow them closely within 1–2 m all day and document their behavior at an individual level, as well as collect recordings of their vocalizations within a distance of 0.5–2 m. Individuals are marked with hair dye or hair cuts to enable easy visual recognition at any time (Jordan, Cherry, & Manser, 2007). Manipulation experiments including presentations of predators or olfactory cues have been conducted to determine the production of calls under controlled conditions, as well as playback experiments to investigate the function of call types and the information receivers may process. For the slender mongooses, yellow mongooses, banded mongooses, and meerkats, genetic analyses are available on intra- and intergroup relatedness. Slender mongooses could only be observed at and around their sleeping site and we only managed to habituate one animal that allowed an observer to follow. Limited experiments close to their sleeping site have been performed. The information on vocal communication for the dwarf mongoose species has been obtained from the literature, mainly from a series of studies by Anne Rasa conducted on a wild group in Kenya in their natural habitat (Beynon & Rasa, 1989; Maier, Rasa, & Scheich, 1983; Rasa, 1986a, 1986b).



3. VOCAL COMPLEXITY IN MONGOOSE SPECIES

The vocal complexity of a species can be viewed as multidimensional, with many different communicative attributes. One obvious dimension is the vocal repertoire size, that is, the number of acoustically discrete call types. Some vocal systems are based on clear discrete calls, whereas other species produce different call types that grade into each other, and it is often difficult to determine the boundaries of the call types (Marler, 1960). Furthermore, many animal species produce discrete vocal units not only singularly but in combination, resulting in sequences of the same units repeated or of different units combined (Kershenbaum et al., submitted). This raises the question, whether a simple description of vocal repertoire size is a reliable measure of the complexity of a vocal system. The complexity of a vocal system may also be associated with external reference, such as functionally referential calls (Marler, Evans, & Hauser, 1992) in comparison to less context-specific calls or relating to the behavior of the caller (Wheeler & Fischer, 2012). Information content may also refer to phenotypic characteristics of the callers, such as body size, status, age, health, condition, and personality traits, which are not necessarily dependent on social or ecological structure. Finally, specific contexts are typically related to specific motivational and

emotional states of the caller and may directly relate to general acoustic structures, as suggested by Morton's motivation-structural (MS) rule hypothesis (Morton, 1977). This rule suggests that a caller's motivation dictates the call structure: an aggressive animal in a hostile situation is predicted to produce a harsh, low frequency sound, whereas a submissive, fearful animal will produce a tonal, high frequency sound.

3.1. Vocal Repertoire Size

Vocal repertoire size has been predicted to correlate with specific social attributes, including group size, group structure, relationship, but also with ecological aspects. Group size in primates (McComb & Semple, 2005) and whales (May-Collado, Agnarsson, & Wartzok, 2007) appears to correlate with the number of discrete call types of a species. Other aspects of social complexity, such as social structure in primates (Gustison et al., 2012), the number of unique social roles in ground-dwelling sciurid species (Pollard & Blumstein, 2012), the mating system in seals (Stirling & Thomas, 2003), or the strength of social bond in primates (McComb & Semple, 2005), seem to also relate to the size of the vocal repertoire. Meerkats have been described as one of the most vocal mammal species (Manser, 1998) including primates, and in a similar way, dwarf mongooses seem to use a large number of different vocalizations (Rasa, 1986a, 1986b). Comparing the different vocal repertoires across the five mongoose species (Table 6.1) shows an increase in the number of discrete call types from the more solitary living mongoose species to the obligate social living species, but not a linear increase with group size. Adult slender mongooses (Beke Graw, unpublished data) and adults in the facultative social yellow mongooses (le Roux, Cherry, & Manser, 2008a, 2008b, 2009) both emit up to eight different discrete call types, whereas adults in the socially obligate species produce many more, in particular, the meerkats (Manser, 1998) and dwarf mongooses (Rasa, 1986a) with more than 30 distinct call types so far described. However, banded mongooses, despite being social and living in groups with the largest number of individuals, only appear to have 15 different call types (Jansen, 2013).

To understand the reasons why we do not find a linear increase with group size in the social obligate species, it helps to assess the exact context of call production. In general, the behavioral repertoire of the mongoose species can be divided into three major contexts: (i) foraging, (ii) antipredator behavior, and (iii) social interactions. This categorization

Table 6.1 Vocal Repertoire of the Different Mongoose Species, Only Call Types Emitted by Adults are Included

Context	Slender Mongoose	Yellow Mongoose	Dwarf Mongoose	Meerkat	Banded Mongoose
Group size	1 (6)	1–13	3–30	3–50	3–70
Call types (total)	>7	>8	>25	>30	>15
Spatial coordination	1	0	5	5	5
Antipredator behavior					
Alarm calls	2	4	>14	12	4
Sentinel calls	0	0	1	6	0
Social interactions					
Affiliative (+)	2	2	2	6	2
Agonistic (–)	2	2	3	3	3

Number of call types in total and depending on the major behavioral context.

makes it immediately obvious why the solitary slender mongooses and the yellow mongooses have few call types as they do not need to coordinate foraging, and likely due to this, also emit fewer alarm calls (le Roux et al., 2008a, 2008b; le Roux, Cherry, Gyga & Manser, 2009). For the social obligate species, the biggest difference exists within antipredator behavior, specifically in the number of different alarm call types and the calls given while on sentinel duty (Table 6.1). Meerkats and dwarf mongooses show a similar alarm call system with predator-type specific alarm calls, whereas the banded mongooses emit more general alarm calls in a graded way along the level of urgency (Furrer & Manser, 2009a, 2009b; Jansen, 2013). Meerkats (Clutton-Brock et al., 1999) and dwarf mongooses (Rasa, 1987) also demonstrate an elaborate sentinel system with accompanying vocalizations, while neither the banded mongooses nor the other two more solitary species show sentinel behavior. Why meerkats use a large number of different sentinel calls (Manser, 1999) and how meaningful the different call types are is an open question. Dwarf mongoose sentinels only emit repeatedly the same call type, which seems to be a slight variation of their close calls (Kern & Radford, 2013; Rasa, 1986a, 1986b), and this has also been described for other species, for example, the cooperative breeding pied babbler (*Turdoides bicolor*) (Bell, Radford, Rose, Wade, & Ridley, 2009).

Comparing the repertoire size of calls given during social interactions across the different mongoose species, the social obligate species have more call types than the more solitary species. Dividing calls into those accompanying affiliative and agonistic interactions shows that solitary and social obligate species use aggressive calls in very similar contexts, such as during aggression over food or in fights, and do not vary much in the number of call types produced. There is, however, a difference between solitary and social species on how many different call types are produced within affiliative contexts, such as grooming, sunning next to each other, and resting. The increase in the number of distinct call types here again is mainly present in the meerkats due to their different call types given while grooming and sunning next to each other (Manser, 1998). Among the less social species, it also appears that the facultatively social yellow mongoose have a large proportion of affiliative vocalizations, likely an indication of this social plasticity, that slender mongoose do not exhibit (le Roux et al., 2009).

A comparative approach describing the vocal repertoire of different mole rat species confirms the pattern found in the mongoose species that group size alone is not a good predictor of vocal repertoire size and other factors must be responsible to explain the variation. While the solitary silvery mole rat (*Heliophobius argenteocinereus*) species shows the lowest number, with eight call types, no correlation is found between-group size and number of call types in the social species (Hrouzkova, 2012). The naked mole rat (*Heterocephalus glaber*), with the largest colonies of 25–204 group members described, seems to produce as few different call types (12) as the other social obligate mole rat species with small group sizes (5–16 group members and 12–18 call types). Both families of mammals, mongooses, and mole rats, support the idea that not group size but other social or ecological attributes are the most important predictors of how many different discrete call types evolve. In line with this, Pollard and Blumstein (2012) suggest attributes that influence vocal repertoire size are more dependent on “what a species has to talk about.” Hence, species with dominance hierarchies, differentiated relationships, coalitions, between-group competition, and mating competition should theoretically have more diverse call systems. Data from the mongoose species seem to support this partly, but more likely the number of behavioral contexts a species discriminates is important, and in particular, in cooperative species displaying labor division and the coordination of these tasks. In addition, the ecological environment and the challenges it imposes seem to play a crucial role. Specifically, within the antipredator context, what options there are to respond in the best way to different predator threats (Macedonia & Evans, 1993) and the necessity to coordinate escape

responses with other group members (Furrer & Manser, 2009a, 2009b), may be driving selective forces.

3.2. Discrete Versus Graded Vocal Systems

The vocal system of a species can be categorized as discrete, graded, or as a mixture of the two (Hauser, 1996; Marler, 1976). Discrete vocal systems are based on discrete vocalizations that are acoustically distinct from each other, and allow receivers an unambiguous discrimination between different call types (Marler, 1976). Graded vocal systems include call types that show a gradual transition from one acoustic structure into another, without clear acoustic boundaries that can be associated with a specific change in meaning (Hauser, 1996; Range & Fischer, 2004). This complicates the already difficult task of classifying vocalizations into biologically meaningful, distinct call types (Keenan et al., 2013).

The evolution of discrete versus graded vocalizations has been argued to depend on the habitat type and precision of information needed in the moment of the production of the signal. Many animal vocalizations are graded, such as the barks in different species including dogs (*Canis lupus familiaris*) (Yin & McCowan, 2004), meerkats (Townsend, Charlton, & Manser, submitted), and primates (Crockford, Herbinger, Vigilant, & Boesch, 2004), and contact calls in chacma baboons (*Papio cynocephalus*) (Fischer, Hammerschmidt, Cheney, & Seyfarth, 2001) and banded mongooses (Jansen, 2013). Graded vocal repertoires have also been suggested to occur in species living in more open habitats with a high rate of interaction with conspecifics at close range (Marler, 1976). Graded signals allow signalers potentially to convey subtle and complex information in their calls about the circumstances they are in. Generally, graded vocalizations have the potential to be rich in subtle information, but there is a trade-off with precision (Green, 1975). While many calls and vocal systems of some animal species, as well as phonemes of human languages, are classified as graded on the production side (Dooling, Brown, Klump, & Okanoya, 1992; Green, 1975; Nelson & Marler, 1990), receivers perceive them as discrete units (Fischer, Metz, Cheney, & Seyfarth, 2001; Hauser, 1996; Marler, 1976; Slocombe & Zuberbühler, 2007). The understanding of a vocal repertoire, and in particular the function of discrete and graded calls, needs to be approached from the production side to determine the contexts of call emission and on the receiver side to identify the extent at which receivers perceive acoustic boundaries and how they categorize graded signals as distinct.

The variation in vocal repertoire size between the social obligate mongoose species seems at least partly due to the fact that meerkats

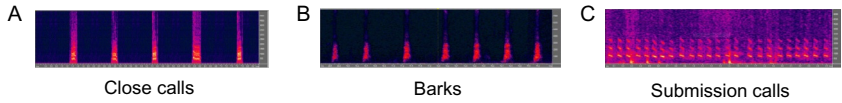
(Manser, 1998) and dwarf mongoose (Beynon & Rasa, 1989; Maier et al., 1983; Rasa, 1986a) have a much more discrete vocal system, while in the banded mongoose, in several contexts, call types grade into each other (Jansen, 2013; Messeri, Masi, Piayya, & Dessifulgheri, 1987). Meerkats (Manser, 2001) and dwarf mongoose (Beynon & Rasa, 1989) evolved predator-type specific alarm calls with a graded continuum along the dimension of urgency, whereas banded mongoose (Jansen, 2013) produce a continuum of alarm calls from low urgency to high urgency without discrimination of the predator type. Similarly, in the context of group coordination while foraging, to maintain group cohesion and avoid foraging competition, meerkats use distinct call types (Bousquet, Sumpter, & Manser, 2011; Manser, 1998; Townsend, Zöttl, & Manser, 2011), whereas banded mongoose use combinations of calls with a graded extension of an additional acoustic element (Jansen, 2013; Jansen, Cant, & Manser, 2012).

3.3. Call Combinations

It has been argued that, due to anatomical constraints, the number of different vocalizations with discrete acoustic structure an animal can produce is limited (Arnold & Zuberbühler, 2006). One way to overcome these limitations is to combine different discrete units into sequences. The combination of discrete acoustic units has the potential to substantially increase the variation in meaningful units with corresponding information available to the receivers. Combinations may happen at the level of combining different elements of discrete acoustic structure within a call type, or the combination of discrete call types into call sequences (Fig. 6.2). Within these two main categories, we can distinguish between call sequences where calls are the simplest discrete acoustic units, and are combined just as repetitions of the same unit, or in fact consist of different acoustic units. The question then rises regarding what aspect the meaning of such call sequences changes in comparison to the single units making up the combination (Collier, Bickel, van Schaik, Manser, & Townsend, submitted; Kershenbaum et al., submitted).

The most commonly described combinations are the syllables in bird and mammalian songs. Syllables are defined as elements of uninterrupted traces in a spectrographic signal and are seldom produced singularly. Syllables by themselves have no meaning and are interchangeable within call sequences (Berwick, Okanoya, Beckers, & Bolhuis, 2011; Marler & Slabbekoorn, 2004). Such combinations are typical in the context of territorial or mate advertisement. However, most of the bird species and also the few mammal

Simple repetitions



Series of multiple call types

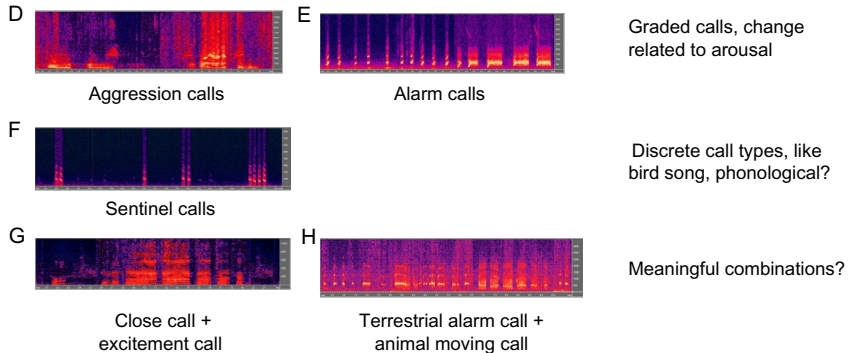


Figure 6.2 Types of call sequences observed in meerkats: (A) close calls; (B) barks; (C) submission calls; (D) aggression calls; (E) terrestrial alarm calls; (F) sentinel calls; (H) terrestrial alarm call combined with animal moving call; and banded mongoose: (G) close call combined with excitement calls.

species producing songs have evolved more structured songs with combinations of discrete elements into distinct note complexes and song phrases, whereby the sequences of the elements are not random and follow strict rules (birds: [Berwick et al., 2011](#); [Nelson & Poesel, 2007](#)); whales: [Ford, 1989](#), [Payne & McVay, 1971](#); rock hyraxes (*Procavia capensis*): [Koren & Geffen, 2009](#); bats: [Bohn, Schmidt-French, Ma, & Pollak, 2008](#)).

Recent evidence for the combinatorial use of discrete vocalizations into meaningful sequences shows the potential continuation from animal vocal communication to human language. The best evidence of combinatorial vocal communication comes from the putty-nosed monkeys (*Cercopithecus nictitans*) and Campbell's monkeys (*Cercopithecus campelli*). Putty-nosed monkeys produce two predator-specific alarm calls referring to aerial or terrestrial predators, and receivers show distinct escape responses to them ([Arnold & Zuberbühler, 2006](#)). In a different context, these two meaningful alarm calls are combined in a specific order, which causes a qualitatively different behavioral response, that is, the group starts to move away ([Arnold & Zuberbühler, 2006, 2008](#)). A different system of call combination is present in Campbell's monkeys, although also in relation to their singularly produced predator-specific aerial and terrestrial alarm calls. These call types also

occur in combination with an acoustic modifier at the end of the call, which causes a different behavioral response in receivers (Ouattara, Lemasson, & Zuberbühler, 2009a, 2009b).

In meerkats and banded mongooses, we have found call sequences being produced in different contexts and also representing different types of sequences (Fig. 6.2). Combinations in their most simple form are just the repetition of the same acoustic units, such as constantly emitted close calls in all three social obligate mongoose species (meerkats: Manser, 1998, Townsend, Hollén, & Manser, 2010; banded mongooses: Jansen, 2013, Jansen et al., 2012; dwarf mongooses: Rasa, 1986a, Sharpe, Joustra, & Cherry, 2010). Also, repeated alarm calls are found within this category, although here, barks or other alarm calls may change acoustically in a graded way in relation to the arousal of the caller, which may be related to the risk or urgency of the situation (Manser, 2001, 2009). However, repetitive call sequences are not restricted to the social obligate mongooses: yellow mongooses emit certain call types (e.g., the peepgrowl and mobbing/recruitment call) in sequences, with the acoustic characteristics of calls changing with varying urgency or motivational levels in the caller. In addition to such simple call sequences, banded mongooses and meerkats combine distinct call types into sequences that appear to convey a different meaning, similar to the combinations described above for primate species.

Banded mongooses constantly produce close calls while foraging, which contain two acoustically different segments within a single syllable call (Jansen et al., 2012). The first segment stays stable and is individually distinct, while the second segment is graded and correlates with the current behavioral context, whether the individual is digging, searching for food or moving. Additionally, in banded mongoose, the identity cue is integrated as a distinct segment within a single call that is independent from context. This first segment likely functions to avoid ambiguity between individuals or receivers having to keep track of several context-specific identity cues. Such a call with two acoustically distinct segments supports Marler's hypothesis of temporal segregation of information by segmental concatenation within a single syllable in nonhuman vocalizations (Marler, 1960), a characteristic commonly seen in human speech (Fitch, 2012). Descriptions of call structures for several other species in the literature suggest that temporal segregation and segmental concatenation of vocal signatures and cues is common, but has so far been neglected as a mechanism of coding information in animal vocal communication (Jansen et al., 2012). Temporal segregation of vocal signatures and cues may evolve in species where communication of

multiple unambiguous signals is crucial, but is limited by the number of call types produced (Jansen et al., 2012).

Banded mongooses also combine discrete call types in other specific contexts while moving around in their habitat and coordinating their activities with their group members. The close call, described above, stays unchanged in these sequences, but is combined with other call types, resulting in longer structures (Jansen, 2013). The addition of an element is observed in three distinct behavioral contexts: when lost from the group, when leading the group, and when encountering rain or moisture. The additional elements differ in the acoustic structure, and also in the number of elements emitted, after the preceding close call. Whether this is correlated to the arousal state of the caller or has another function still needs to be tested. However, it shows that banded mongooses appear to use nonrandom combinations of calls into more complex sequences also within affiliative contexts, and not only in alarm calls where so far most call combinations have been described.

Meerkats also produce combinations out of two meaningful call types into a new and different meaningful call type (Manser, Hollén, Steinert, Dewas, & Townsend, in preparation; Townsend, Charlton, & Manser, in preparation). Meerkats emit terrestrial alarm calls to predators on the ground (Manser, 2001), and they produce a specific call type, the animal moving call, when either an animal on the ground passes by or a bird flies close to ground, but also in response to stationary animals that subtly move body parts (Manser et al., in preparation). In certain contexts, specifically when terrestrial predators are moving, meerkats combine predator-specific terrestrial alarm calls with animal moving calls into longer call sequences. Receivers seem to respond to these naturally occurring sequences by increasing their antipredator behavior in comparison to the two call types produced by themselves (Townsend & Manser, in preparation). When playing back the two different call types by themselves in comparison to their combination, meerkats took longer to relax when exposed to the potentially more threatening terrestrial + animal moving call combination than when exposed to either of the meaningful calls by themselves or any of the other artificially created combinations or their individual call components (Townsend & Manser, in preparation). This suggests that meerkats can extract meaningful compositional information from combinations of referential calls.

The examples of different combinations of call types in meerkats and banded mongooses show the potential to increase flexibility and meaning of acoustic signals when vocal repertoires are limited. In particular, banded mongooses with only few distinct call types in their vocal repertoire seem to

combine these call units in many different ways to form new call types that have different meanings. How meaningful they are, however, still needs to be tested with playback experiments. The work on meerkats with a detailed description of the production of such call combinations and the perception tested with playback experiments, however, shows the potential for such combinations in animal vocal communication, and indicates them as potentially analogous to combinatorial sequences described for human language (Collier et al., submitted; Townsend & Manser, in preparation). It is likely also that the dwarf mongooses show similar ways of combining their distinct call types (Rasa, 1986a, 1986b). We can expect that in the future, such call combinations will be discovered for many more bird and mammal species, besides what so far has mainly been described for primate species (Fitch & Zuberbühler, 2013).

3.4. Functionally Referential Calls Versus Less Context-Specific Calls

Animals produce discrete call types that are highly context specific, while other call types are given in several different contexts. Few animal species have been described to emit unique call types to specific external events or objects in the environment of the caller, where the receivers also show distinct behavioral responses to these calls (for recent reviews, see Gill & Bierema, 2013; Townsend & Manser, 2013). These call types have been referred to as functionally referential (Macedonia & Evans, 1993).

Meerkats show one of the most sophisticated alarm call systems described for any animal species with predator-type specific and more general alarm calls (Manser, 2009; Manser, Seyfarth, & Cheney, 2002). The predator-type specific calls also vary in their acoustic structure depending on the risk of the predator and its distance, taken together to form a composite measure of the urgency of the situation (Manser, 2001; Manser, Bell, & Fletcher, 2001). Furthermore, meerkats produce a call referring to moving animals in their surrounding independent of their risk (Manser, 2010; Manser et al., in preparation), whereby they combine this call type in sequences with the predator-type specific calls in dangerous situations. Dwarf mongooses also use many discrete alarm call types that vary with predator type and other predator traits (Beynon & Rasa, 1989). Banded mongooses, neither yellow nor slender mongooses, appear not to have these functionally referential alarm calls, but show a much more graded system. The evolution of functionally referential calls has been explained by the need to have immediate precise information on the context that allows the receiver to show the

appropriate response without knowledge of the rest of the context (Manser, 2013; Meise, Keller, Cowlshaw, & Fischer, 2011).

The underlying mechanisms in the production of and the response to functionally referential calls have been suggested to be cognitively less demanding than those driving less context-specific calls (Wheeler & Fischer, 2012). The functionally referential aspect could be based on very simple associative mechanisms on the production and perception side, whereby, for example, predator-type specific alarm calls seem to have all the necessary information included to enable the appropriate behavioral response. Differently, for less context-specific calls, at least on the perception side, a receiver cannot fully rely on just the acoustic structure of the call and needs additional information to show the appropriate response. However, several examples show that those species that produce and respond to functionally referential calls do not follow an inflexible pattern in responding. Rather, receivers often take additional information of the general context at that moment of hearing the call into account when responding.

An example demonstrating the flexibility of receivers to respond to functionally referential calls comes from variation in responses based on foraging investment. Meerkats that have already invested substantial time digging for a specific prey item at the same location are less likely to immediately run for shelter than the same individual to the same call when it is just foraging on the surface (Amsler, 2007). This pattern is also supported from work in primates, where receivers are more likely to respond to functionally referential calls of reliable callers than to other less reliable callers (Cheney & Seyfarth, 1988). Meerkats need to immediately respond to urgent aerial alarm calls to run to the closest shelter, as a raptor approaches very fast (Manser et al., 2001). Shelters, typically boltholes, separated approximate every 45 m from each other (Manser & Bell, 2004), can be located in any direction of foraging meerkats in the moment of alarm calls. Yet, in 80% of the time, meerkats have been shown to run to the closest shelter, independent of whether they had just passed that location or if the closest bolthole was in a direction they had not passed (Manser & Bell, 2004). These examples clearly show that receivers, when they hear a functionally referential alarm call, integrate different types of information and, according to that information, show flexible responses. Therefore, the claim that more information processing is involved in less context-specific calls (Wheeler & Fischer, 2012) may not, overall, be valid.

Another aspect that supports the claim that functionally referential calls may not be simple genetically determined calls is the fact that these calls

develop at a later age than nonreferential calls in similar contexts. Meerkats already produce general alarm calls at an early age when they start foraging with their groups (Hollén & Manser, 2006, 2007; Hollén, Clutton-Brock, & Manser, 2008). However, they only begin to produce the predator-type specific alarm calls several months later, at the age of about 6 months (Hollén & Manser, 2007). This suggests that a young individual needs to develop the association between the external stimuli and also to learn to integrate the surrounding context (Graw & Manser, 2007; Hollén & Manser, 2006). Furthermore, meerkats do not produce medium urgency predator-specific alarm calls when they are by themselves (Townsend, Allen, & Manser, 2012; Townsend, Rasmussen, Clutton-Brock, & Manser, 2012), indicating they have control on the production side, and not only on the perception side, as to how to respond (Manser & Bell, 2004). These findings suggest that the acoustic structure of call types in most mammals may be rather inflexible, but the timing of call production and response decisions are seldom simple associative reflexes, but are based on information processes with multiple inputs (Manser & Fletcher, 2004; Seyfarth & Cheney, 2003).

3.5. Information on Individual Attributes

The hypothesis positing the relationship between social complexity and communicative complexity predicts for social groups with individualized social interactions that group size correlates with information related to individuality. The argument is that the more group members that are present to interact the more they may profit from being able to distinguish among group members. Empirical evidence on this aspect comes from several studies including infant distress calls in bats (Wilkinson, 2003), birds (Beecher, 1990), and sciurids (Blumstein & Armitage, 1998; Pollard & Blumstein, 2012).

In meerkats and banded mongooses, many call types are individually distinct in their acoustic structure. However, while some of these individual differences seem important in the social interactions, in other contexts they seem not to be used by receivers. In meerkats, alarm calls, sentinel calls, and close calls have been investigated for individual differences in the acoustic structure. While in all these tested call types on the production side, we find clear differences for specific acoustic parameters, receivers seem not to attend to individual variation in alarm calls. When foreign alarm calls are played back, receivers respond in the same way as if one of their group members

had called (Manser et al., 2001; Schibler & Manser, 2007; Townsend et al., 2011). Habituation-dishabituation playback experiments confirmed that they do not discriminate between individual callers (Schibler & Manser, 2007). In close calls, we found similar individual variation confirmed for meerkats (Townsend et al., 2010) and banded mongooses (Müller & Manser, 2008), and with playback experiments, we showed that receivers use this information to socially monitor the spatial position of other group members (Müller & Manser, 2008; Reber, Townsend, & Manser, 2013; Townsend et al., 2011). However, the playback experiments of close calls in meerkats also showed that overt behavioral responses often only occur during conflict situation, with no visible responses occurring outside this context (Reber et al., 2013). This emphasizes the need to be careful in the interpretation of negative results, where no response is shown, as it is not possible to determine whether receivers have not perceived the differences, whether it is not meaningful in that context, or whether they are not motivated to respond in the tested situation (Jansen, Cant, & Manser, 2012).

Vocalizations can also relate to nonmorphometric individual traits, such as status, age, condition, and the emotional state of a caller in the moment of producing the signal. In meerkats, behaviorally different contexts are related to subtle differences in the acoustic structure of the calls. Meerkats that interrupt foraging for a short time and stand up to scan for predators often emit a close call as they change from a bipedal standing position to a quadrupedal walking position. Acoustic analysis of these “guarding” close calls in comparison to “foraging” close calls showed an acoustic difference (Townsend et al., 2011). When testing the response of foraging animals to these two acoustically very similar close call types, subjects decreased their own vigilance when hearing guarding close calls in comparison to being played foraging close calls. Similarly, in some of the alarm calls we observed slight variations in the acoustic structure within the same alarm call type in form of nonlinearities (Townsend et al., 2011). Despite these calls sounding very similar, meerkats respond stronger to the calls with the subharmonic structure than alarm calls lacking it (Townsend & Manser, 2011). Testing the potential function of these nonlinearities in habituation experiments suggests that these structures prevent receivers to habituate to the same call type (Karp, Manser, Wiley, & Townsend, 2014). These examples show how important very subtle differences in the acoustic structure can be in the organization of different tasks.

Not only do the vocalizations of adult individuals convey information regarding various attributes of the caller, but individual differences in vocalizations also play an important role in young animals, which often have their own vocal repertoire and only start to produce adult vocalizations when they have become independent from adult provisioning. Vocalizations, in particular, play an important role in competition between siblings to get the attention of their parents, maintain contact or to be fed or protected by them. In the banded mongoose, dependent pups form long-lasting and exclusive associations with particular adults, their “escorts,” and receive the majority of care from these individuals. Playback experiments showed that both pups and escorts were more responsive to calls of their association partners than to calls of other individuals (Müller & Manser, 2008). These results suggest that pups and escorts recognize each other vocally and mutually and that both pups and providers contribute to the maintenance of the pup-escort associations. Pups may benefit from vocal recognition of their escorts since this reduces the time spent alone, vulnerable to predators and without being fed. Escorts may be more responsive to their associated pup’s calls than to another pup’s calls because they preferentially care for this particular individual. In meerkats, pups are fed by all adults when they forage with the group from the age of 4–12 weeks on (Brotherton et al., 2001), and here individual differences in their vocalizations seem to play a less important role than the advertisement of hunger level. Pups constantly emit several different types of begging calls (Kunc, Madden, & Manser, 2007; Manser & Avey, 2000), which vary in call rate with hunger (Manser, Madden, Kunc, English, & Clutton-Brock, 2008) and the acoustic structure varies with age (Madden, Kunc, English, & Clutton-Brock, 2009a, 2009b). Adult meerkats respond to the differences in the vocal behavior of the pups and provide more food to pups calling at a higher rate (Carlson et al., 2006; Manser et al., 2008), but also adjust the type of prey items they provision to the age of the pups, which they seem to perceive via the acoustics of the most frequently emitted begging call type (Thornton & McAuliffe, 2006).

3.6. Expression of Emotions

Animal vocalizations have for a long time been considered to relate to the motivational state (Morton, 1977) of a caller and directly express the emotion of it (Darwin, 1872). This view also included the assumption that animals had little control over the production of a call and the structure of the

call type was directly linked to the emotional state of the caller related to the specific context it experienced. The notion that animals have no control over the production of a call has been proven to be wrong, as many species adjust their production of calls in the presence of an audience (Fichtel & Manser, 2011; Karakashian, Gyger, & Marler, 1988; le Roux et al., 2008a; Townsend, Allen, et al., 2012; Townsend, Rasmussen, et al., 2012; Zuberbühler, 2008). Yet, the general patterns put forward by Morton (1977) as motivational structural rule and how they vary depending on the motivational and emotional state of the caller have found some empirical support in birds (Morton, 1977) and mammals (August & Anderson, 1987). However, there are also studies that point out that some other species produce sounds that clearly do not follow these rules (Blumstein & Chi, 2012; Branch & Freeberg, 2012).

Meerkats and yellow mongooses have both been shown to adjust their vocal behavior to the social environment, indicating that vocal production is at least partly controlled by the caller (le Roux et al., 2008a; Townsend, Allen, et al., 2012; Townsend, Rasmussen, et al., 2012). In meerkats, males solitarily prospecting for reproductive opportunities are less likely to produce alarm calls compared with when they are with conspecifics, either in coalitionary prospecting groups or in their home group (Townsend, Allen, et al., 2012; Townsend, Rasmussen, et al., 2012). Furthermore, the probability to emit alarm calls increased with the urgency of the predation risk, indicating that variation in alarm call production depends on whom the call is addressed to, as well as the motivational state of the caller. Similarly, in the facultative social yellow mongooses, solitary foraging individuals do not emit alarm calls, but do warn conspecifics with alarm calls when they are nearby (le Roux et al., 2008a, 2008b).

There is evidence from the different mongoose species that the acoustic structure of calls follows Morton's motivation-structural rules and relates to specific behavioral context reflecting the motivational and emotional state of a caller. Morton suggested that bird and mammal vocalizations in aggressive or hostile contexts (aggressive sounds) should be low frequency and noisy, and vocalizations in fearful or appeasement contexts should be high frequency and tonal (Morton, 1977). The vocal repertoires of the different mongoose species confirm the general pattern that has been described for 50 mammal species in that the main aggressive and fearful situations are reflected in the predicted acoustic structure by the MS rules, but as shown for other bird and mammal species (August & Anderson, 1987), the acoustic

structure of some alarm calls emitted to more general contexts are not fully explained by these rules (Manser, 2009).

There is some evidence from meerkat, banded mongoose and yellow mongoose calls that the emotional state of a caller does not fully explain the acoustic structure of calls in given situations. First, the animal moving call in meerkats is given in several contexts related to different emotional states of the caller, and no simple model expressing the arousal of a caller could explain the consistent acoustic structure of this call (Manser, 2010). The animal moving call can be given in a relaxed context (encounter of herbivores or nondangerous birds), as well as in highly stressful predator encounters (with dogs or raptors) (Manser et al., in preparation), and yet its acoustic structure is very similar in these different situations. This suggests a different underlying mechanism than the expression of just arousal and an additional dimension of emotions, and if it was an expression of emotions, some control from the caller on which call to produce must be present. In a similar direction, banded mongoose's constantly emitted close calls are given in combinations with other calls that relate to high arousal situations, such as being lost from the group or excited about a resource (Jansen, 2013). This confirms that calls related to rather different emotional states can be given in close sequences. Whether this is just an expression of an ambivalent context, however, is not clear so far. Finally, yellow mongooses emit the peepgrowl during aggressive interactions with group members, but also as a repetitive alarm. Yet, both contexts (aggression and alarm) are polar opposites, according to Morton's MS rules. These examples raise the question whether emotional expressions can be regarded as simple processes or whether they can involve some more complicated underlying mechanisms as well.



4. CONCLUSIONS AND FUTURE DIRECTIONS

Vocal complexity in meerkats and other mongoose species shows how variable vocal communication can be, even in phylogenetically closely related species. Differences in vocal repertoires, whether vocal systems are discrete or more graded, how combinations of discrete calls are used in call sequences, and the context-specific and individual attributes of calls, all emphasize that we need to know an animal species or system very well to understand what the large variation in vocalizations in animals means, and what selective factors favor different acoustic attributes to vary. For the different mongoose species, social and ecological factors affect the way of

communication in different ways, both when to call and how the signals look like.

The social environment clearly affects the evolution of communication. However, which specific attributes of social complexity relate to which attributes of communicative complexity is not easy to determine as general patterns. While the theoretical framework has been laid out, we are still far away from being able to use a comparative approach on many different taxonomic families to test the hypotheses. This is certainly due to the lack of the same or similar data being collected in different species. Meerkats and other mongooses, as described here, provide a unique opportunity to address the questions in phylogenetically close species. Despite research on these species over many years, we lack much of the data in all the different species. Furthermore, data collection is a slow process. We need to coordinate our data collection much more in systematic ways to answer the many questions raised here in view of a comparative approach.

To understand the variation and complexity of communication in animal systems, we need to identify attributes of the social and ecological environment and how these relate to specific communicative attributes. For vocal communication, this includes identifying specific acoustic parameters varying with social attributes, as well as personality traits, hormones, and morphometric aspects. Studying the ontogeny of calling behavior will not only help us to understand the function but also to identify the underlying cognitive mechanisms of the production as well as of the perception of vocalizations. Within the comparative approach of different species, we need to account for morphometric differences in their vocal tract as well as their hearing abilities (psychophysics) (Ramsier, Cunningham, Finneran, & Dominy, 2012), the habitat they live in, and how this may affect the transmission of call types. Only such a comprehensive approach will help us to understand the variation of communication even within one modality.

This review on the research done on vocal communication on the different *Herpestidae* species shows a huge variation and diversity of signals, partly related to social aspects of the species, partly related to ecological differences. Much insight on questions related to different attributes of communicative complexity, including vocal repertoire size, graded versus discrete vocal system, individual based differences, and the expressions on emotions has been gained. Species like those of the mongoose family offer a great opportunity, as they allow habituation to a stage that enables detailed behavioral observations, as well as recordings and playback experiments in their natural habitat.

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